Read the assignment carefully. Clearly identify what is all asked of us. Define the scope of the question, and find an appropriate solution. The solution shouldn't be overly complicated. Prioritize easy to understand basic code which is longer and uses our basic simple tools to short very complex code calling very niche functions and using unnecessarily complicated libraries.

Based on the above, come up with an implementation plan to solve the entire exercise. Verify that your implementation meets all the requirements. After that, give me a detailed solution of the entire assignment, along with modular code and theory answers. The code must be complete, don't leave filler spaces like "to be done" . It should also have human sounding comments which explain what the code is doing. Implement the entire code yourself. Along with the assignment solution also give me precise instructions on what each part of the code does and well as how to run the code on my Mac. I have venv to run the code.

It is important that you give detailed instructions on how to run the code, adn get the results. Not just "this is how to run a python script", but rather stepswise instructions eg:"these commands to load the dtaa, then this to run the code, then u get results then u do xyz…." I dont want to put your generated results in my final submission, rather I want to understand the code you have given, run it on my device, get the results, get the theory answers and your results interpretation to make a final report myself.